Microsoft

FinOps toolkit SRE Agent

Manage your Azure estate — capacity and cost — across the lifecycle.

FinOps toolkit community release training.

Built by **FinOps Toolkit**

In partnership with **Azure SRE Agent product team.**

One agent, one estate, one lifecycle

The agent runs against the customer's existing FinOps hub.

Six lifecycle stages: Forecast → Procure → Allocate → Deploy → Operate → Optimize / decommission.

At every stage, capacity and cost are reasoned about together.

5 specialist subagents · 3 skills · 33 tools · 18 scheduled tasks.

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Today's agenda — walk the lifecycle

Stage 1 — Forecast
Stage 2 — Procure
Stage 3 — Allocate
Stage 4 — Deploy (the pre-deploy gate)
Stage 5 — Operate
Stage 6 — Optimize / decommission
Stage 7 — Improve the agent itself

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This lifecycle, the FinOps Framework, and your team

FinOps Framework phase →	Forecast	Procure	Allocate	Deploy	Operate	Optimize
FinOps Inform	**Primary**	*in scope*	**Primary**	*in scope*	*in scope*	*in scope*
FinOps Optimize	*in scope*	**Primary**	*in scope*	*in scope*	*in scope*	**Primary**
FinOps Operate	*in scope*	*in scope*	*in scope*	**Primary**	**Primary**	*in scope*

Primary — the FinOps Framework phase that drives the work at this lifecycle stage.

in scope — phase still applies, but it isn't the primary lens at this stage. Every cell is in play.

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Stage 1 — Forecast

CAPACITY

will we have quota / SKU / zone capacity for projected growth in the next 90 days?

COST

what will it cost, and where do current commitments sit?

project both sides forward; flag where supply or budget breaks first.

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Worked example — a workload growth scenario

Capacity

- agent runs sku-availability lookup, vm-quota-usage, zone mapping check; flags 60-day quota burn at current trajectory.

Cost

- agent runs cost-forecasting-model, commitment-discount-utilization; projects savings plan coverage drift if growth lands without re-commit.

Combined answer: paired risk score with two recommended actions — file quota request now, plan SP top-up Q3.

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Scheduled tasks at this stage

Capacity tasks

capacity-quarterly-strategy	Q
storage-paas-growth-forecast	M
compute-utilization-trend	W

Cost tasks

ytd-report	Q
mom-report	M

The chief-financial-officer specialist narrates variance against forecast at each cadence.

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What the chief-financial-officer specialist does in Forecast

Translates the agent's projections into variance language: "forecast vs budget," "variance to plan," "confidence band."

Surfaces the assumptions the projection rests on (utilization growth rate, commitment expiry dates, regional expansion).

Calls out which figures are stable vs provisional.

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Why Forecast matters

Capacity outcome

supply risks surfaced 60–90 days before they block deployment.

Cost outcome

commitment drift caught before it costs a quarter of premium-rate spend.

Lifecycle-loop outcome

forecast feeds the Procure stage's quota and commitment decisions.

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Stage 2 — Procure

CAPACITY

should we request quota, buy capacity reservations, move regions?

COST

should we extend savings plan / RI commitments?

every supply procurement decision pairs with a commitment decision.

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Worked example — a CRG with paired RI

Capacity

- agent recommends a capacity reservation group for the target SKU + region + zone.

Cost

- agent identifies that pairing the CRG with a 3-year reservation drops the rate ~60% versus on-demand.

Combined answer: one recommendation — procure CRG and pair with RI; flag that the reservation must be applied to the same scope.

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Scheduled tasks at this stage

Capacity tasks

capacity-monthly-planning M

Cost tasks

benefit-recommendation-review M

cost-optimization M

Procurement runs on a monthly cadence to catch every commitment cycle.

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What the chief-financial-officer specialist does in Procure

Frames every procurement recommendation in commitment-rate-impact terms.

Compares the recommendation against the customer's commitment portfolio.

Quantifies expected savings against execution risk for board-ready language.

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Why Procure matters

Capacity outcome

capacity reservations procured for guaranteed supply.

Cost outcome

commitment posture stays current with workload growth.

Lifecycle-loop outcome

procured capacity feeds the Allocate stage's distribution decisions.

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Stage 3 — Allocate

CAPACITY

is quota distributed correctly across subs and quota groups? Are zones aligned? Are CRGs assigned to the right workloads?

COST

are tags correct so allocation rolls up cleanly?

between procurement and deployment, the inventory needs to land in the right places.

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Worked example — zone mapping drift across two subs

- Customer shares a CRG across two subscriptions for failover.
- Agent runs zone mapping check; detects that subscription A's logical zone 2 maps to physical zone X, but subscription B's logical zone 2 maps to physical zone Y.
- The CRG can't actually serve both — silent reliability risk.
- Agent flags the drift; recommends explicit zone-pinning.

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Scheduled tasks at this stage

Capacity tasks

non-compute-quota-audit W

sku-availability-audit W

Cost tasks

no shipped task addresses tag and allocation hygiene at this stage today

The cost-side gap is one PR away — see Stage 7.

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Why Allocate matters

Capacity outcome

capacity actually lands where the workload runs.

Cost outcome

spend rolls up to the right team / product / customer.

Lifecycle-loop outcome

clean allocation makes the Deploy gate trustworthy.

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Stage 4 — Deploy (the pre-deploy gate)

CAPACITY

can we use it right now — quota, SKU, zone, CRG attachment?

COST

what's the budget impact and the anomaly risk?

the gate produces evidence and a verdict; humans execute the deployment.

v1 = read + recommend + report. Pipeline integration is a v2 conversation.

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The 8-check pre-deploy gate

Check	Side
Quota headroom	Capacity
SKU availability	Capacity
Zone alignment	Capacity
CRG attachment	Capacity
Capacity signal freshness	Capacity
Budget impact	Cost
Commitment coverage	Cost
Anomaly baseline	Cost

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Worked example — a gate run that returns 7 / 8 green

- Customer asks the agent: "can we deploy this VMSS in eastus2 today?"
- Agent runs the 8 checks in parallel.
- Returns: 7 green, 1 yellow on capacity signal freshness (last sku-availability-audit was 5 days ago in a region with active capacity events).
- Recommendation: re-run sku-availability-audit before deploying; gate verdict pending.

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Scheduled tasks at this stage

Capacity tasks

capacity-daily-monitor D

Cost tasks

budget-coverage-audit M

alert-coverage-audit M

The daily monitor pre-stages capacity evidence so the gate has fresh signals when called.

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Why Deploy matters

Capacity outcome

deployment failures from quota / SKU / zone gaps caught before resources change.

Cost outcome

budget breaches and anomaly drift caught before they compound.

Lifecycle-loop outcome

clean deployments feed the Operate stage's monitoring with predictable baselines.

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What the agent does NOT do (yet)

The agent does not file quota increase requests.

The agent does not gate the pipeline directly — it produces gate evidence; you wire it in.

The agent does not manage Quota Groups boundaries today (audit only).

Each of these is a v2 conversation.

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Stage 5 — Operate

CAPACITY

how is it being used — quota burn, CRG utilization, drift?

COST

what's the spend trend, anomalies, AI cost, allocation gaps?

the agent runs the recurring rhythm so your team operates the estate, not the dashboard.

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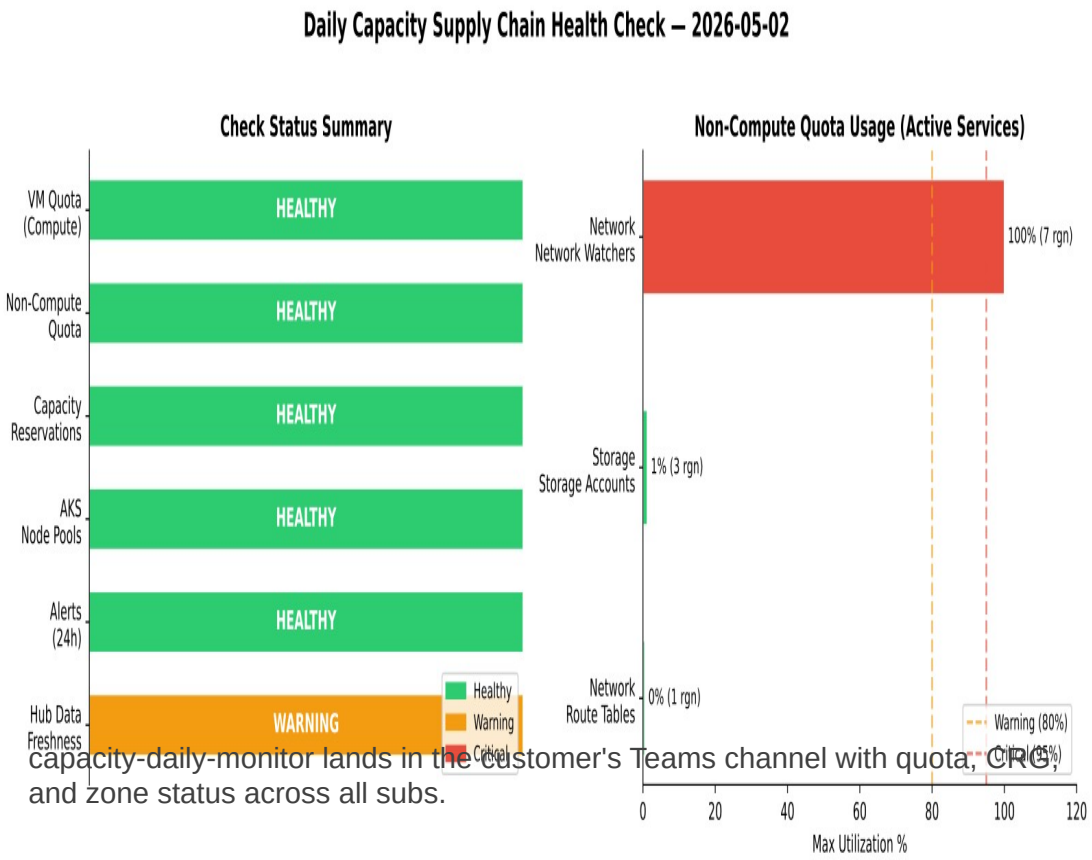
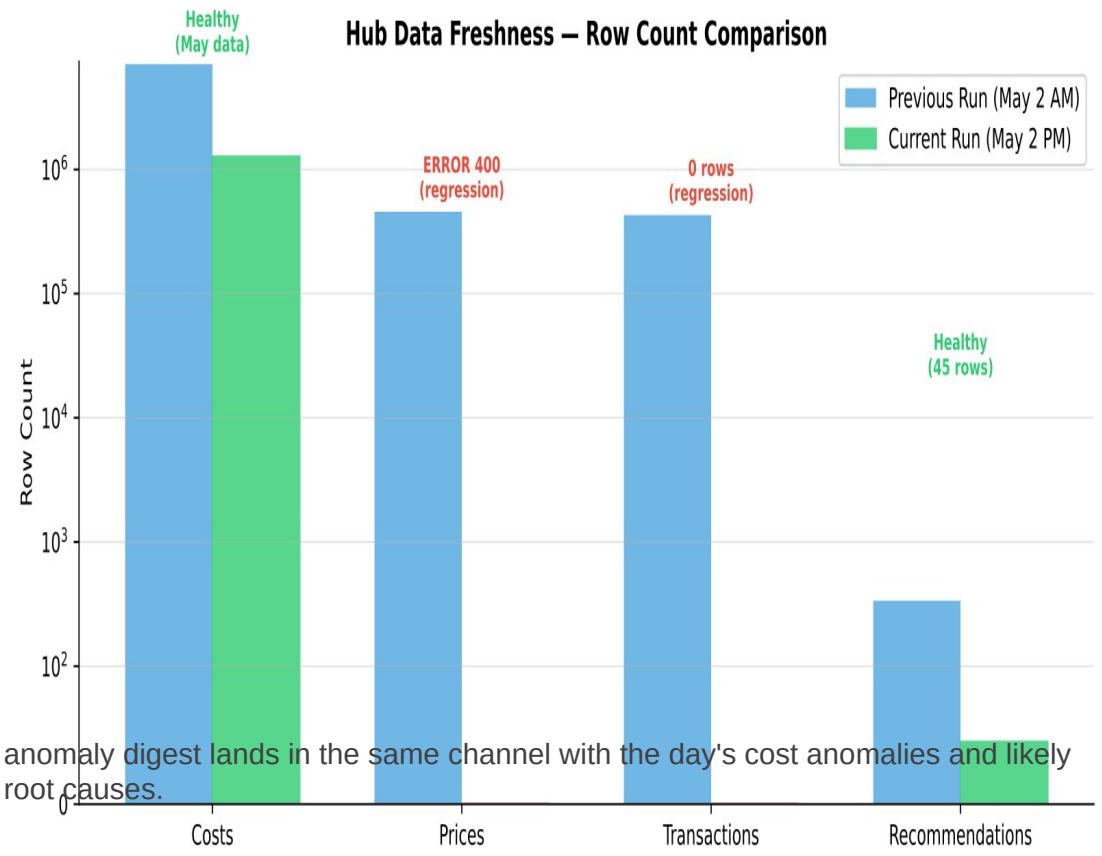
Worked example — daily cadence (capacity + cost)

Capacity



capacity-daily-monitor lands in the customer's Teams channel with quota, CRG, and zone status across all subs.

Cost



anomaly digest lands in the same channel with the day's cost anomalies and likely root causes.

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Worked example — weekly + AI cost

- Weekly: capacity-weekly-supply-review and budget-coverage-audit deliver to the same channel.
- Monthly: ai-workload-cost-analysis surfaces token-volume drivers, model-substitution opportunities, prompt optimization targets.
- Customer's AI spend story stops being "why did this jump" and starts being "here's what changed and what to do."

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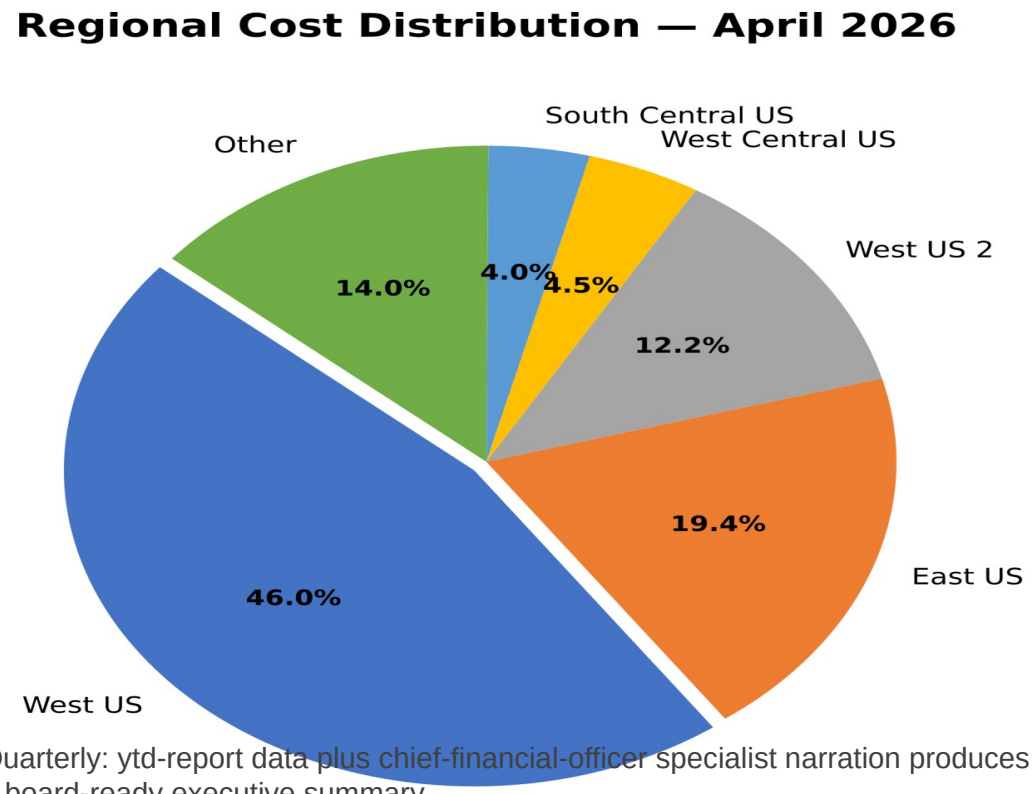
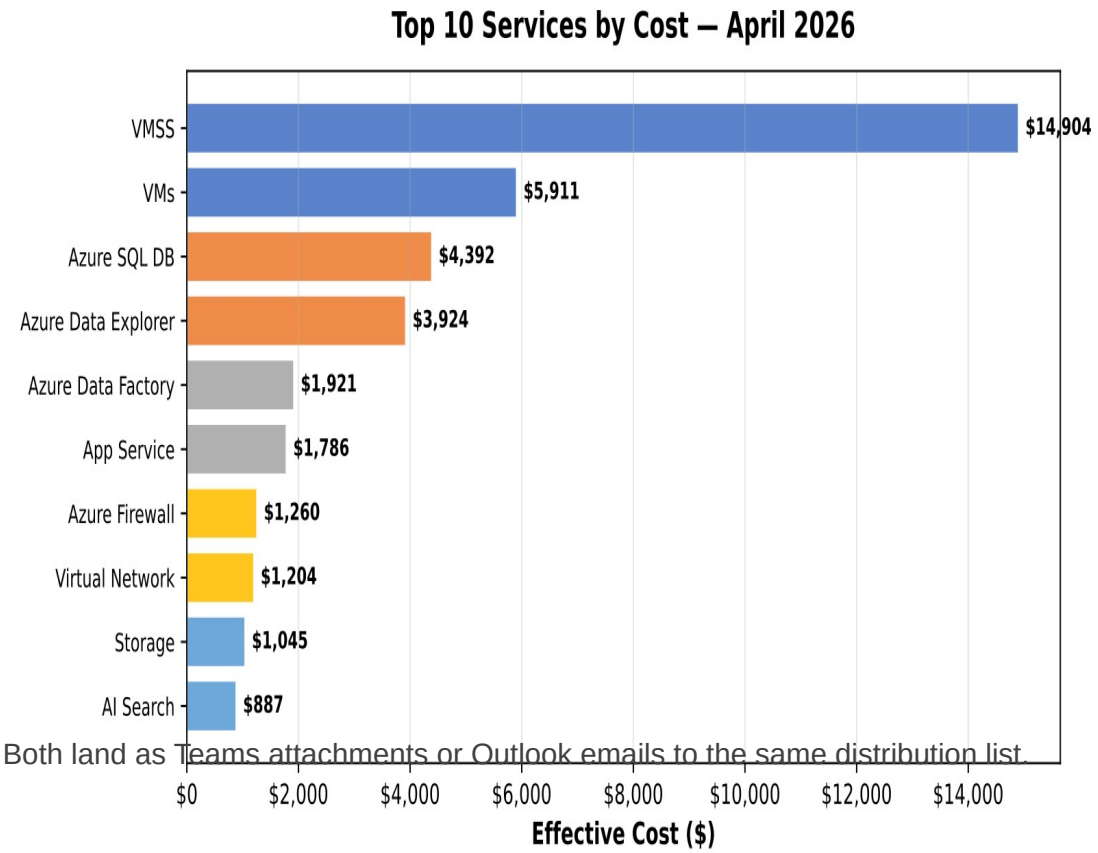
Worked example — quarterly executive narrative





Quarterly: ytd-report data plus chief-financial-officer specialist narration produces a board-ready executive summary.
Capacity-quarterly-strategy delivers the supply-side equivalent.

Both land as Teams attachments or Outlook emails to the same distribution list.

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Scheduled tasks at this stage

Capacity tasks

capacity-daily-monitor	D
capacity-weekly-supply-review	W
monitoring-scope-validation	M

Cost tasks

ai-workload-cost-analysis	M
advisor-suppression-review	M
hubs-health-check	D

Hubs health check fires daily so every other report inherits a freshness verdict.

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Why Operate matters

Capacity outcome

drift caught before it becomes outage.

Cost outcome

anomalies caught before they compound; AI cost trends visible weekly.

Lifecycle-loop outcome

operate signals trigger Optimize / decommission decisions when usage patterns shift.

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Stage 6 — Optimize / decommission

CAPACITY

can we release it — idle capacity, surrendered reservations, detached CRGs?

COST

can we recover the spend — stranded commitments, decommission cost, AI workloads winding down?

where the lifecycle loops back to Forecast.

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Worked example — a workload winding down

- Customer asks: "if we decommission this AI workload over the next two months, what happens?"
- Agent runs the projection: idle capacity to release (compute quota + AOAI quota), reservations that lose coverage, CRG attachment to drop.
- Combined answer: a release plan with capacity targets, commitment surrender / rebalance options, and the cost recovery profile.

Combined answer: a release plan with capacity targets, commitment surrender / rebalance options, and the cost recovery profile.

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Scheduled tasks at this stage

Capacity tasks

no shipped task addresses idle-capacity-release detection today

Cost tasks

no shipped task addresses stranded-commitment detection today

Both gaps are one PR away — see Stage 7.

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Why Optimize matters

Capacity outcome

idle quota and stranded CRGs returned to the pool.

Cost outcome

commitments rebalanced; decommission cost surfaced before invoice surprise.

Lifecycle-loop outcome

optimize signals trigger Forecast re-baseline; the cycle closes.

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Stage 7 — Improve the agent itself

Pull requests, not feature requests.

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Two audiences, two loops, one repo

For platform engineers

pull requests, not feature requests — your YAML, your KQL, your scheduled tasks, version-controlled.

For FinOps practitioners

your queries, not their dashboards — KQL against your hub, tuned to your allocation model, instead of waiting for a vendor report.

Same loop, same fork, two audience hooks.

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Gaps you found in this lifecycle, as PRs your team can ship

Stage 3 (Allocate) cost-side

tag and allocation hygiene scheduled task.

Stage 3 (Allocate) capacity-side

zone-mapping-validation scheduled task.

Stage 6 (Optimize) capacity-side

idle-capacity-release scheduled task.

Stage 6 (Optimize) cost-side

stranded-commitment-detection scheduled task.

Each one is a YAML file, a Kusto query, and a PR.

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Four structural moats

Data sovereignty — financial data stays in the customer's tenant.

Customization without vendor roadmap — PRs against version-controlled YAML.

No vendor lock-in — FOCUS for data, KQL for queries, YAML for config, Teams for delivery.

Capacity and cost reasoned about together from the same agent with the same data trust model.

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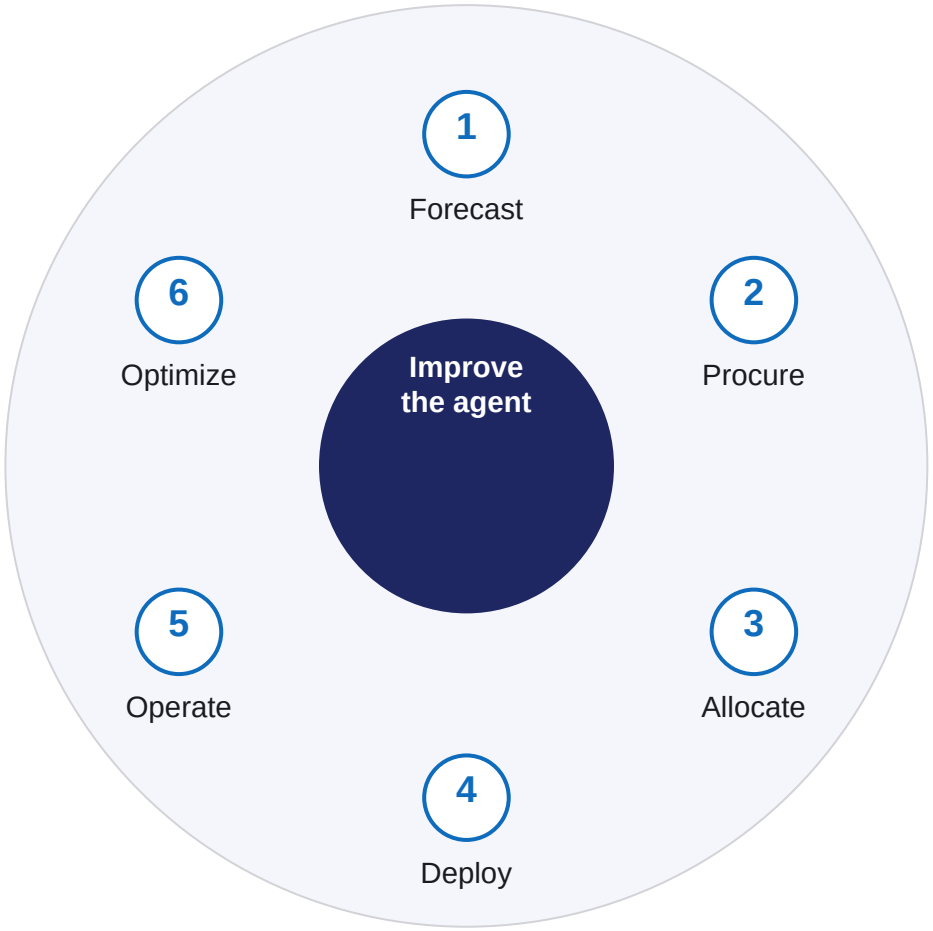
The lifecycle as one wheel



One agent, one estate, one continuous loop.

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Deploy and learn more

Deploy: azd up against an existing FinOps hub; post-provision srectl applies skills, agents, tools, knowledge, scheduled tasks.

Pricing: Azure SRE Agent runtime + Azure consumption (ADX, LAW, AppInsights) + ~0.25–0.5 FTE; license $0; payback under 12 months at any meaningful estate scale (full breakdown in appendix).

Docs: aka.ms/ftk/sre-agent. Source: github.com/microsoft/finops-toolkit. Issues: GitHub label Tool: SRE Agent. Internal: FTK community Teams channel.

Bring back what breaks.

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Architecture, briefly

Customer prompt or schedule

Teams · CLI · cron

Agent in customer subscription

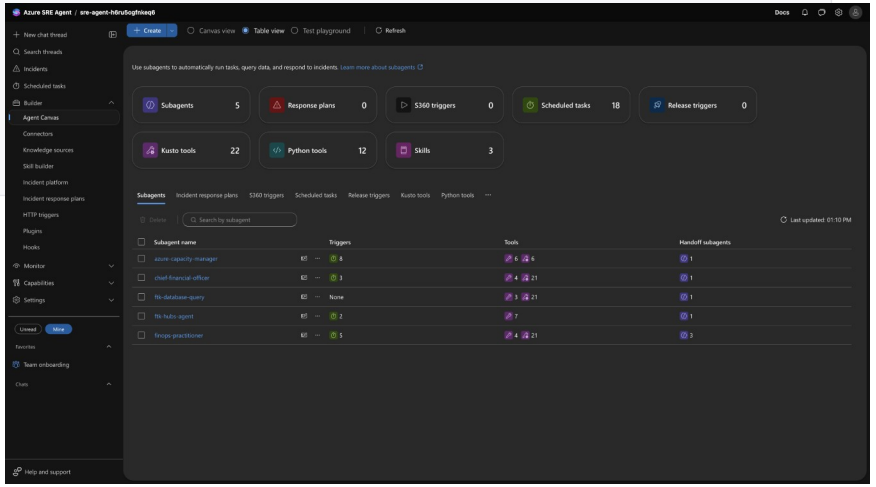
5 specialists · 33 tools · 18 tasks

Two data sources

FinOps hub ADX · Azure control plane

RBAC · v1 read-only

Deploy: subscription-scoped Bicep + post-provision srectl. RBAC: managed identity + Reader + Monitoring Contributor + optional ADX AllDatabasesViewer + custom role for zone peers.



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TCO comparison at scale

Approach	Detail
Reference scenario	$6B annual Azure estate.
CloudHealth (VMware/Broadcom)	$300K–$600K/yr SaaS license + $50K–$150K implementation.
Cloudability (Apptio/IBM)	$250K–$500K/yr + comparable implementation.
Spot.io (NetApp)	$200K–$400K/yr + comparable implementation.
FinOps toolkit SRE Agent	$87K–$206K/yr (hub infra + ADX + agent + ~0.25 FTE) + ~$50K–$80K partner stand-up.
Net annual savings	$100K–$400K plus data sovereignty, customization, no vendor lock-in.

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Cost breakdown by component

Component	Cost basis
Azure SRE Agent runtime	~$300/mo fixed per agent + agentic-unit consumption (variable).
Azure Data Explorer cluster	standard meters, depends on cluster SKU and ingestion volume.
Log Analytics workspace	ingestion + retention.
Application Insights	ingestion + retention (linked to Log Analytics).
Storage account	minimal.
Operations FTE	~0.25–0.5 (varies by customer maturity).
Open source license cost	$0.

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Sizing scenarios by estate

Estate band	Annual cost & payback
Small estate (~$50M annual Azure spend)	minimal ADX footprint, ~$15K–$30K/yr total agent + infra; payback under 6 months on FinOps practitioner time alone.
Mid estate (~$500M)	moderate ADX, ~$40K–$90K/yr total; payback under 6 months including 3–5 point commitment utilization lift.
Large estate ($5B+)	full ADX cluster, ~$87K–$206K/yr total; payback under 12 months including 5–10 point commitment utilization lift and recovered FinOps capacity (1–2 FTE equivalent).

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